NARESH i technologies

Advanced

Gen AI & Agentic AI



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Generative AI & Agentic AI

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Course Modules

Module 1: Python Programming

- **Python Installation & Setup**
  - Python Installation
  - Package Distributions — Anaconda, Miniconda
  - Python Package Managers — pip, conda
  - Python Environments



- **Data Types**
  - Basic Data Types — int, float, str, bool
  - Collection Data Types — list, tuple, set, dict
- **Control Flow**
  - Conditional Statements — if, elif, else
  - Loops — for, while
- **Modules, Packages, and Libraries**
  - Understanding Modules, Packages, and Libraries
  - Importing in Python
  - from and import
- **Functions**
- **Higher Order Functions**
  - Map
  - Filter
- **Anonymous Function**
  - Lambda
- **Decorators**
- **Generator**
  - Yield vs Return

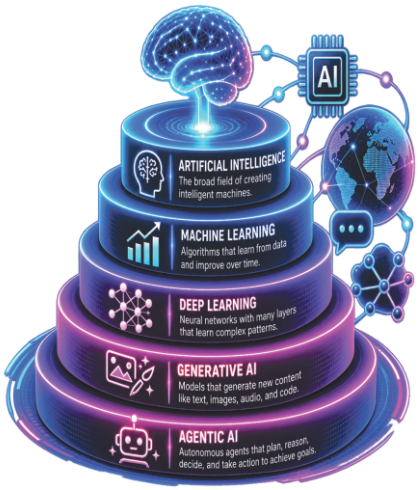


- **File Handling**
- **OOP — Object-Oriented Programming**
  - Class
  - Data Members
  - Inheritance
  - Encapsulation
- **Data Engineering Modules**
  - Numpy
  - Pandas
    - Series
    - DataFrame

Module 2:
Deep Learning Foundations

Neural Networks

- Introduction to Neural Networks
- Mathematical intuition behind Neural Networks
- Neurons and Layers
- Weights & Biases
- Activation Functions
- Forward & Backward Propagation
- Loss Functions & Optimization



PyTorch

- Tensors
- nn.Module
- Dataset and DataLoader
- Training the Model
- TensorBoard — Loss Tracking
- GPU – SM Cores, Cache and VRAM

Key Concepts of NLP

- Why vectors?
- Corpus
- Tokenization –
  - BPE – Byte Pair Encoding
  - WordPiece,
  - SentencePiece
- Embedding
- Vocabulary
- Attention vs Self-Attention



Engineering Ecosystem

- **Basics of APIs & JSON**
  - What is an API?
  - Making API Requests
  - Understanding JSON Format
- Docker
- Git Vs Github
- IDEs – VS Code, Google Colab

Module 3:
Fundamentals of Generative AI & LLMs

Introduction

- What is Generative AI?
- What is Agentic AI?
- Generative AI vs Agentic AI
- What are LLMs?

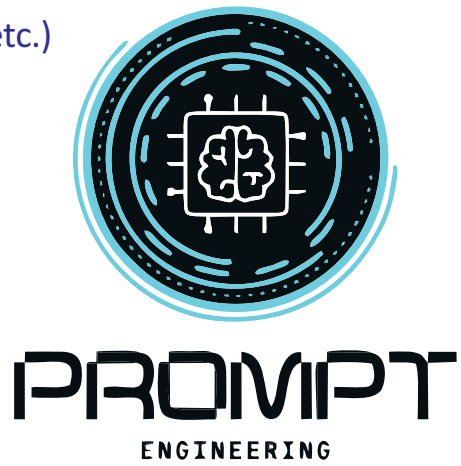


Transformer Architecture

- Self-Attention Mechanism
- Positional Encoding
- Contextual Embedding
- Multi-head Attention
- Cross Attention
- Encoder–Decoder

Prompt Engineering

- **Prompts**
  - What is Prompt?
  - Prompts Vs Instructions
  - Role, Task, Input, Instruction, Context, Output
  - Max Sequence Length vs Max Output Tokens
  - Task-specific Prompts
- **Sampling Parameters** (Temperature, Top-k, Top-p, etc.)
- **Prompting Techniques**
  - Zero-Shot Prompting
  - Few-Shot Prompting
  - Chain of Thought (CoT)
  - ReAct
  - Self –Consistency
- **Guardrails**



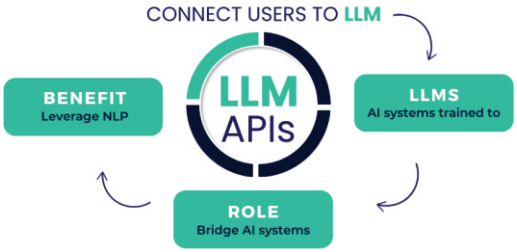
Module 4:
Working with LLMs

- Open Source Vs Closed Source LLMs
- **HuggingFace Ecosystem**
  - Models, Datasets and Spaces
  - Model Loading
  - Model Parameters
  - Model Weight Formats (.pth, Safetensors, ONNX)
  - Model Size Calculation and Licence
- **Hugging Face Libraries**
  - Transformers
  - Trainer
  - PEFT
  - Tokenizers
  - Datasets
- Multi-modal LLMs: Text, Audio (ASR, TTS), Image, Video



LLM APIs & SDKs:

- **LLM API Fundamentals**
  - API vs SDK
  - Authentication & API Keys
  - Request / Response
  - Messages and Roles
  - System, User and AI/Assistant Messages
- **LLM Provider SDKs**
  - OpenAI SDK
  - Google Gemini SDK
  - Anthropic SDK
- **LLM Generation Interfaces**
  - OpenAI chat.completions.create()
  - OpenAI responses.create()
  - Using **OpenAI-compatible interface** to access Gemini models
  - Gemini generate_content()
  - Anthropic messages.create()



Module 5: Retrieval-Augmented Generation (RAG)

- What is RAG?
- LLM Hallucination: Causes and Mitigation
- When to Use RAG
- **Components of RAG:**
  - Embedding models
  - Vector Databases: Chroma, Pinecone, FAISS, Milvus
  - Chunking Strategies

- Conversational RAG
- Embedding Spaces: Semantic Similarity & Cosine Distance
- Answer Grading / Response Evaluation

- **RAG Evaluation**
  - RAGAS Framework
  - Retrieval Evaluation
  - Answer Evaluation
  - Faithfulness
  - Relevance
  - Context Precision
  - Context Recall

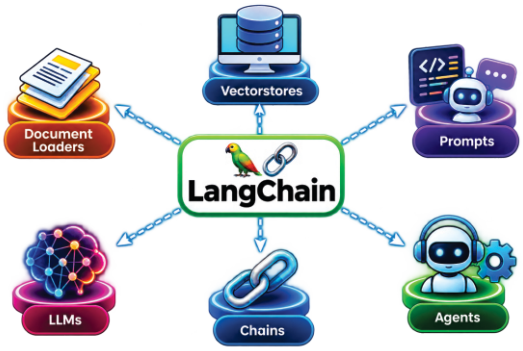


Advanced RAG Patterns

- Corrective RAG (CRAG)
- Self-RAG with Reflection
- Graph RAG
- Hybrid RAG (Semantic + Keyword)
- Re-ranking

Graph RAG

- Graph Fundamentals
  - Nodes
  - Edges
  - Properties
  - Relationships
- Ontology Design
- **Graph Databases**
  - Neo4j
  - Community vs Enterprise vs Cloud
  - Cypher Query Language (Cypher)
    - CREATE
    - MATCH
    - SET/ REMOVE
    - DELETE
  - Graph Queries
  - GraphRAG - RAG Implementation using Graph



Module 6: LangChain Framework

- Introduction to LangChain
- Prompt Templates, ChatPromptTemplates
- **Chains**
  - Sequential Chain

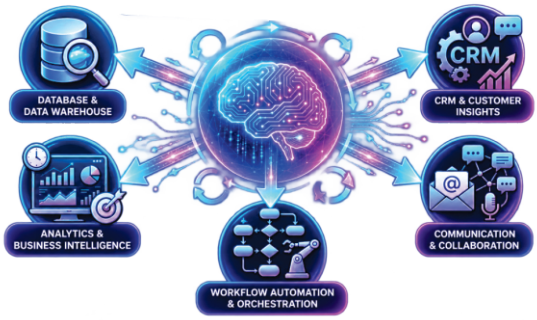
- Parallel Chain
- Conditional Chain

- Structured Output and Output Parsers
- **Building RAG Pipelines with LangChain**
  - Basic Document Loaders and Text Splitters
  - Retrievers, VectorStore
- Tool Calling
- Agents

Module 07: Agentic AI Principles

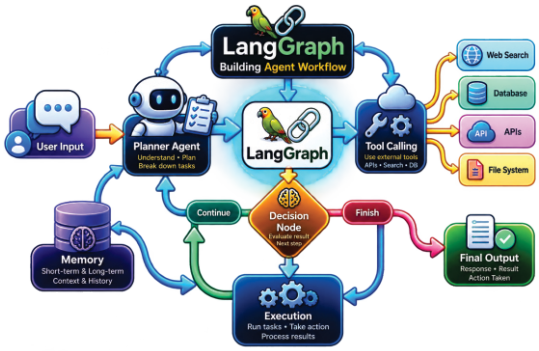
Agentic AI Principles

- What is Agentic AI?
- AI Agents Vs Agentic AI
- **Agentic AI Frameworks Overview**
  - No-code vs Code-first
  - n8n vs LangGraph vs Airflow
- **Design Principles:**
  - Goal, Planner, Orchestrator
  - Human in the Loop
- **Agentic AI Frameworks**
  - CrewAI, LangGraph



LangGraph — Building Agent Workflows

- Agentic AI using LangGraph
- LangChain vs LangGraph
- **LangGraph Components**
  - State
  - StateGraph
  - Nodes
  - Edges
  - Conditional Edges
- **Workflow Types:**
  - Sequential
  - Parallel
  - Routing
  - Orchestrator-Worker
  - Evaluator
- **State & Memory Management**
  - Persistence Strategies
  - Short term memory
  - Long term memory

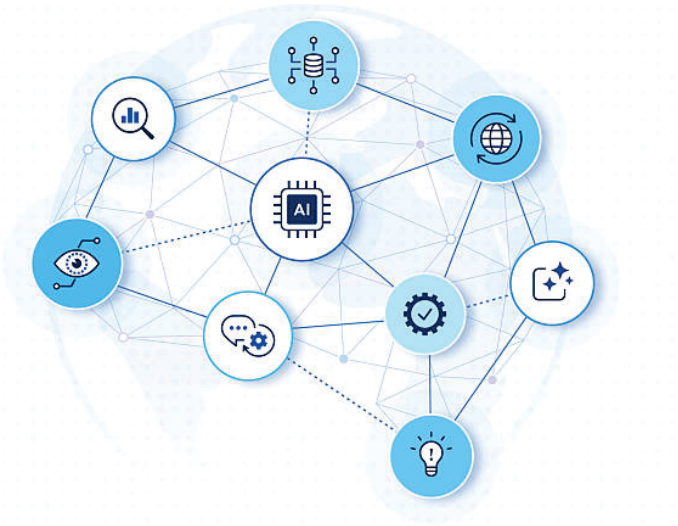


- Checkpoints
- Threads

- **Advanced LangGraph**
  - Time Travel in LangGraph
  - Interrupts
  - Human in the Loop
  - Streaming – Values & Updates
  - SubGraphs

CrewAI

- Intro to CrewAI
- What is a Crew?
  - Agents
  - Tasks
  - Process
- Manager–Worker
- Peer-to-Peer



Advanced Agentic AI Topics

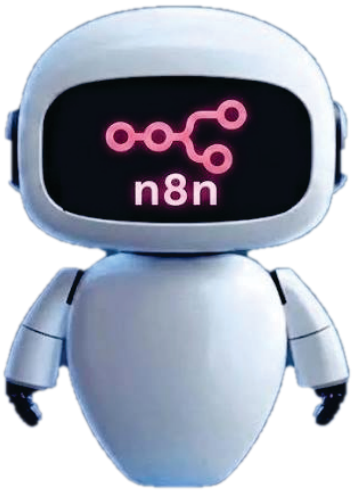
- Context Engineering
- Harness Engineering
- Loop Engineering
- Skills
- Middleware & Hooks

Module 08: n8n

- **Introduction to n8n**
  - What is n8n?
  - Features and Use Cases
  - Self-hosted vs Cloud
  - Comparison with Other Automation Tools (Zapier, Make)
- **Installation of n8n**
  - Docker Installation
  - npm Installation
- **n8n Interface**
  - Editor UI
  - Canvas
  - Node Panel
  - Credentials
  - Executions
  - Templates
  - Projects and Folders



- **Understanding Workflows**
  - Creating Workflows
  - Executing Workflows
  - Manual vs Triggered Execution
- **n8n Nodes**
  - **Trigger Nodes**
    - Manual Trigger
    - Schedule Trigger
    - Webhook Trigger
    - Chat Trigger
  - **Action Nodes**
    - HTTP Request
    - Set Node
  - **Logic Nodes**
    - **IF**
    - **Switch**
- **Creating Agentic Workflows**
  - Understanding AI Agents
  - Building an AI Agent
  - Agent with Memory
  - Agent with Tools
- **Integrating External Services**
  - Gmail
  - Telegram/LinkedIn



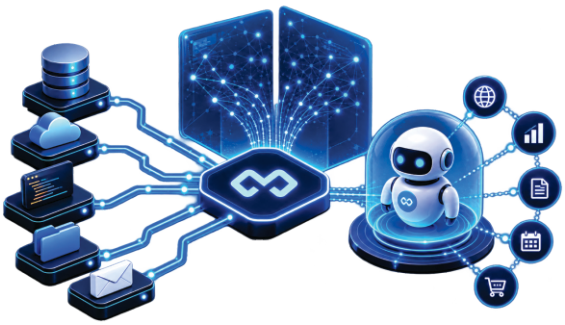
Module 09: Observability with LangSmith

- **Instrumentation**
- **Telemetry**
- **Observability**
- **What is Observability?**
  - Metrics
  - Logs
  - Traces
- **LLM Observability**
  - Latency
  - Token Usage
  - Cost
  - Time to First Token
  - Throughput
  - Concurrency



Module 10: Model Context Protocol (MCP)

- MCP Fundamentals and Architecture
- **MCP Server**
  - MCP Local Server
  - MCP Remote Server
- MCP Client
- MCP Host
- FastMCP and LangChain MCP Adapter
- Claude Desktop
- Tool Integration through MCP
- MCP vs Traditional Integration Patterns
- GITHUB MCP integration



Module 11: Ollama & Local AI

- **Introduction to Ollama**
  - What is Ollama?
  - Features and Benefits
- **Using Ollama**
  - Deploying Models Locally
- **Accessing Models**
  - REST API
  - Python Library
  - LangChain Integration
- **Customizing Models**
  - Introduction to the Modelfile
  - Creating Custom Models

Module 12: Fine-Tuning and Quantization

- Training stages – Pre-Training Vs Fine-Tuning
- When to Use Fine-tuning Vs Prompt Engineering Vs RAG
- Fine-Tuning Approaches
  - Unsupervised / Self-Supervised
  - Supervised Fine tuning(SFT)
  - RLHF
- **Parameter-Efficient Fine-Tuning (PEFT)**
  - LoRA
  - QLoRA
  - Adapters
  - Rank
- **Model Evaluation**

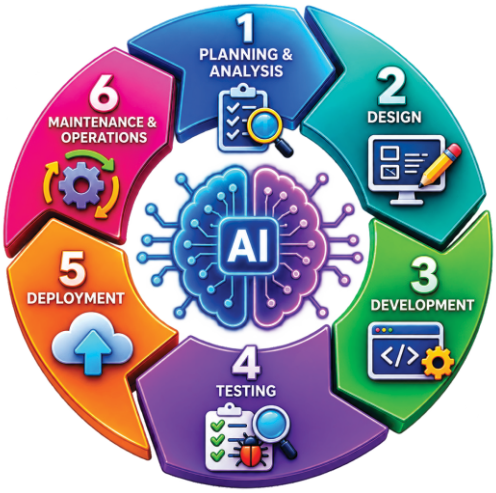


- Training Loss
- Validation Loss
- Overfitting Vs Underfitting

- **Quantization Techniques**
  - Intro to Quantization
  - Asymmetric vs Symmetric
  - Post-training Quantization Vs Quantization-Aware

Module 13: Model Serving & LLM Deployment

- **Inference Optimization**
  - KV Cache
  - Continuous Batching
  - Paged attention
  - Speculative Decoding
  - Flash Attention
- **Model Serving Frameworks**
  - Inference Engines
    - vLLM
    - TensorRT-LLM
  - Inference Servers
    - vLLM Server
    - Triton Inference Server



Module 14: AI Driven - SDLC

- **Code Assistants**
  - Claude Code
  - GitHub Copilot
  - Cursor
- **GitHub Spec Kit**
  - Specification-Driven Development
  - Constitution
  - Spec → Plan → Tasks → Implementation
  - AI-assisted software development workflow

*** Generative AI & Agentic AI Certification Guidance ***

Capstone Projects

Each module includes hands-on mini-projects. The following capstone projects are designed to integrate and apply concepts from multiple modules.

1. **Domain-Aware LLM Chatbot Using Open-Source and API-Based Models**
   Build a versatile chatbot that supports multiple knowledge domains by switching between open-source models (e.g., Mistral) and API-based models (e.g., OpenAI GPT).

2. **Customer Support Assistant Powered by Retrieval-Augmented Generation (RAG)**
   Develop a support automation system that uses vector search and LLMs to deliver accurate, context-aware responses from internal knowledge bases.

3. **Agentic AI Advisor for Healthcare (or Domain-Specific) Guidance and Decision Support**
   Create an intelligent advisor using multi-agent orchestration and memory, capable of assisting users with goal-driven tasks in a regulated or specialized domain.

4. **Stock Market Insight Generation Using Agentic AI and Web Search Tools**
   Build an agentic AI system that autonomously gathers, analyzes, and summarizes real-time stock market news and trends using web search tools, and provides actionable stock suggestions.



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